

IMO Training – Inequalities

Level 1 Assignment 1

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Both paper and electronic copies are accepted.
- Deadline for submission: 24 August, 2019

1. Show that $\forall a, b, c \in \mathbb{R}^+, \sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{c^2 + a^2} \geq (a + b + c)\sqrt{2}$.
2. Show that $\forall a, b, c \in \mathbb{R}^+$ s.t. $a^2 + b^2 + c^2 = 1$, $\frac{a^3}{b} + \frac{b^3}{c} + \frac{c^3}{a} \geq 1$.
3. Show that if $a > b > 0$, $a + \frac{1}{ab - b^2} \geq 3$.
4. Show that $\forall a_1, a_2, \dots, a_n \in \mathbb{R}^+$ s.t. $a_1 + a_2 + \dots + a_n = 1$,

$$\left(a_1 + \frac{1}{a_1}\right)^2 + \left(a_2 + \frac{1}{a_2}\right)^2 + \dots + \left(a_n + \frac{1}{a_n}\right)^2 \geq \frac{(n^2 + 1)^2}{n}.$$

5. If $a, b, c \in \mathbb{R}^+$ s.t. $a^3 b^2 c = 1$, find the minimum of $a + b + c$.
6. If $x, y \in \mathbb{R}^+$, prove that $x^8 + y^4 \geq 4x^2 y - 2$.
7. If $0 < a, b, c < 1$ and $a + b + c = 2$, prove that $8(1 - a)(1 - b)(1 - c) \leq abc$.
8. If $a, b, c, d > 0$ s.t. $c^2 + d^2 = (a^2 + b^2)^3$, prove that $\frac{a^3}{c} + \frac{b^3}{d} \geq 1$.
9. Show that $\forall a, b, c, d > 0$,

$$\frac{a^3 + b^3 + c^3}{a + b + c} + \frac{a^3 + b^3 + d^3}{a + b + d} + \frac{a^3 + c^3 + d^3}{a + c + d} + \frac{b^3 + c^3 + d^3}{b + c + d} \geq a^2 + b^2 + c^2 + d^2.$$

10. * If a, b, c are the sidelengths of a triangle, show that $a^2 b(a - b) + b^2 c(b - c) + c^2 a(c - a) \geq 0$.
11. * Show that $\forall a, b, c \in \mathbb{R}_{\geq 0}$ s.t. $a + b + c = 1$, $7(ab + bc + ca) \leq 2 + 9abc$.

Classwork

3.1 Inequality Q3, 7, 15, 17, 19, 20, 23, 25, 27, 28, 30, 31, 32, 34, 44, 47, 50.